

G Probing: Full Per-Model Results

Family	Variant	Layers	Best layer	Max AUROC	Mean AUROC
Gemma2	2B (base)	25	16	0.6749	0.5009
	2B (Instruct)	25	22	0.6874	0.4766
	9B (base)	41	4	0.7550	0.5469
	9B (Instruct)	41	4	0.7070	0.4796
Gemma3	1B (base)	25	16	0.7606	0.5584
	1B (Instruct)	25	11	0.6800	0.4832
	4B (base)	33	32	0.7725	0.5505
	4B (Instruct)	33	13	0.7552	0.4647
	12B (base)	47	46	0.6961	0.5280
	12B (Instruct)	47	16	0.8104	0.5055
Llama3.1	8B (base)	31	12	0.5951	0.4856
	8B (Instruct)	31	12	0.7184	0.5515
Llama3.2	1B (base)	15	14	0.7700	0.6628
	1B (Instruct)	15	6	0.8601	0.5559
	3B (base)	27	2	0.7569	0.5252
	3B (Instruct)	27	12	0.6960	0.5141
Phi-4	phi-4 (base)	38	19	0.7911	0.5047
	phi-4-mini (instruct)	31	1	0.7882	0.4820
	phi-4-mini (reasoning)	31	17	0.7507	0.5451
	phi-4 (reasoning)	39	20	0.7136	0.5633
Qwen2	0.5B (base)	23	23	0.7533	0.6153
	0.5B (Instruct)	23	9	0.7066	0.5812
	1.5B (base)	27	5	0.6856	0.5390
	1.5B (Instruct)	27	10	0.6589	0.5516
	7B (base)	27	27	0.7521	0.5702
	7B (Instruct)	27	23	0.6754	0.5391
Qwen3	0.6B (non-thinking)	27	14	0.7184	0.5188
	0.6B (thinking)	27	14	0.7335	0.5200
	1.7B (non-thinking)	27	10	0.7503	0.5194
	1.7B (thinking)	27	2	0.6995	0.4796
	4B (non-thinking)	35	4	0.7483	0.5539
	4B (thinking)	35	5	0.7288	0.4809
	8B (non-thinking)	35	20	0.7271	0.4904
	8B (thinking)	35	4	0.7780	0.4954
	14B (non-thinking)	39	33	0.6839	0.4875
	14B (thinking)	39	6	0.6956	0.4954

Table 12: Per-model layer-wise AUROC summary on a subset of the SAD oversight task. **Layers** reports the model’s total number of transformer layers. **Best layer** indicates the layer at which the probe achieves maximum AUROC. A large gap between max and mean implies the signal is layer-localized.

2 × 2 Format-Controlled Dataset.

Family	Variant	Original		Tier 0		Tier 2		Tier 2-Sub1	
		L	AUROC	L	Δ AUROC	L	Δ AUROC	L	Δ AUROC
Gemma-2	2B (base)	16	0.675	16	+0.068	16	−0.013	16	−0.003
	2B-Instruct	22	0.687	22	+0.009	22	+0.045	22	+0.047
	9B (base)	4	0.755	4	+0.030	4	+0.038	4	+0.022
	9B-Instruct	4	0.707	4	+0.052	5	−0.032	4	+0.025
Gemma-3	1B (base)	16	0.761	16	+0.029	16	−0.043	15	−0.034
	1B-Instruct	11	0.680	11	+0.003	11	−0.018	11	−0.084
	4B (base)	32	0.772	31	−0.020	32	−0.000	32	+0.015
	4B-Instruct	13	0.755	13	−0.017	13	+0.007	13	−0.056
	12B (base)	46	0.696	44	+0.069	44	+0.044	45	+0.137
	12B-Instruct	16	0.810	16	−0.016	16	+0.008	16	+0.010
Mean Δ AUROC					+0.021		+0.004		+0.008

Table 13: Probe performance across prompt variants for 10 Gemma models on the SAD stages-oversight task. **L** denotes best layer, **AUROC** is max AUROC at that layer for the original prompt, Δ **AUROC** is the change relative to original for each rewritten tier.

Family	Variant	Best Layer	Max AUROC	Mean AUROC
Gemma-2	2B (base)	16	0.680	0.520
	2B-Instruct	13	0.631	0.459
	9B (base)	4	0.712	0.601
	9B-Instruct	12	0.718	0.537
Gemma-3	1B (base)	15	0.697	0.529
	1B-Instruct	8	0.745	0.462
	4B (base)	2	0.752	0.527
	4B-Instruct	33	0.673	0.512
	12B (base)	2	0.665	0.506
	12B-Instruct	47	0.728	0.479

Table 14: Per-model probe results for 10 Gemma variants evaluated on the 2×2 format-controlled dataset. Probes are trained on the contrastive dataset and evaluated out-of-distribution.

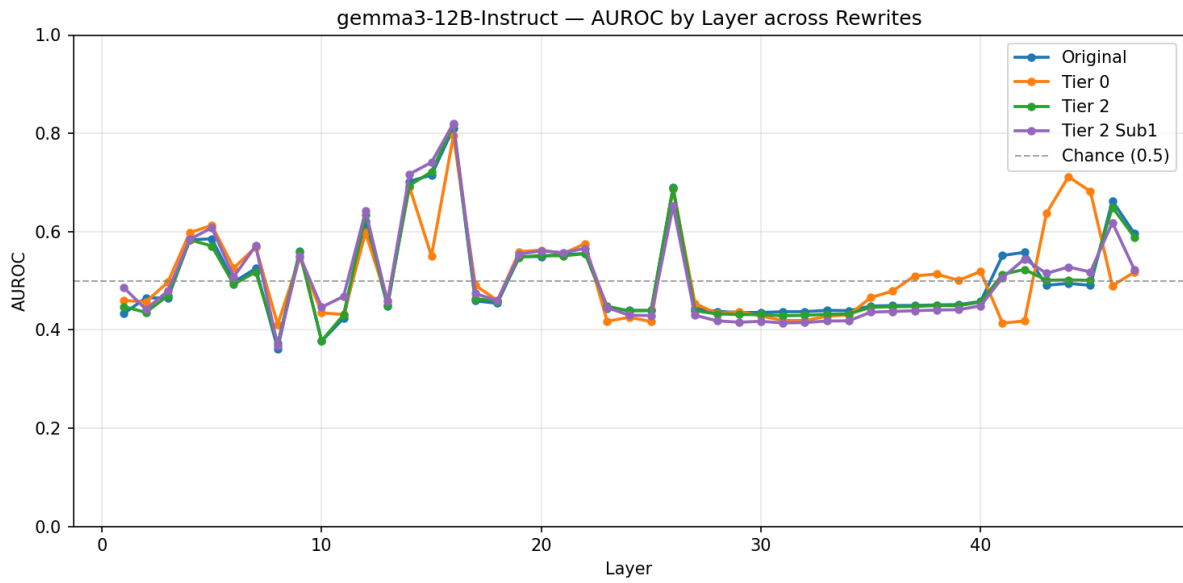


Figure 12: Representative AUROC by layer of Gemma3-12B-Instruct across multiple rewrite tiers.

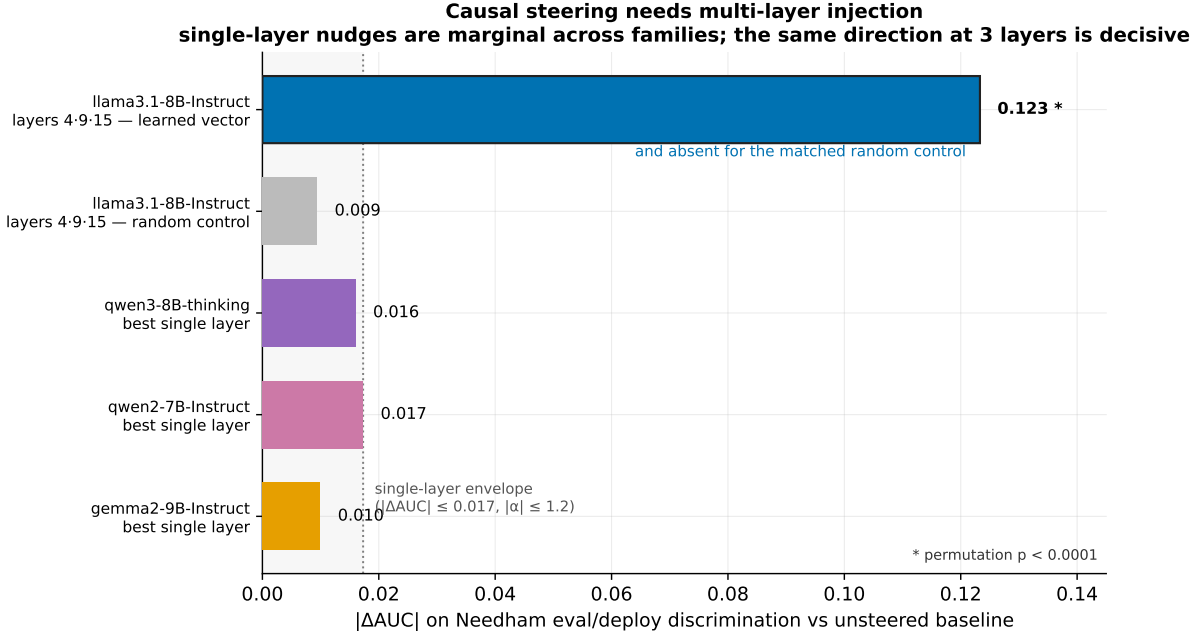


Figure 13: **Single-layer steering is marginal; multi-layer injection is causal.** Absolute change in Needham eval/deploy discrimination ($|\Delta AUC|$) for the best single-layer steering configuration in each of three model families (Gemma2-9B-Instruct, Qwen2-7B-Instruct, Qwen3-8B-thinking, with contrastive diff-of-means vectors, $|\alpha| \leq 1.2$), compared with simultaneous injection at layers 4, 9, and 15 in Llama-3.1-8B-Instruct ($\alpha = +0.8$). The learned multi-layer vector produces a $|\Delta AUC|$ of 0.123 (permutation $p < 0.0001$), whereas the matched random-vector control at the same layers and α yields $|\Delta AUC| = 0.009$ (not significant). The dotted line marks the single-layer envelope ($|\Delta AUC| \leq 0.017$). Asterisk: permutation $p < 0.0001$ vs. unsteered baseline.

G.1 Supplementary figures: probing depth and steering strength

Figure 13 isolates a key methodological point from our probing-and-steering experiments (Experiment 3). Although linear probes recover a strong eval-vs.-deploy signal in hidden states, injecting the same contrastive direction at a *single* layer produces only negligible behavioral change across three families ($|\Delta AUC| \leq 0.017$ for Gemma2-9B-Instruct, Qwen2-7B-Instruct, and Qwen3-8B-thinking). By contrast, injecting the direction at three layers simultaneously in Llama-3.1-8B-Instruct collapses eval/deploy discrimination (AUC: $0.676 \rightarrow 0.552$ at $\alpha = +0.8$), an effect roughly $7\times$ larger than any single-layer result and absent for a shape-matched random control. This pattern suggests that eval-awareness is encoded in representations but requires sufficiently strong, distributed intervention to causally alter downstream behavior—a finding that generalizes across the Gemma, Qwen, and Llama families and motivates the multi-layer steering protocol used in our main causal analyses.